

InTraScale: Accelerating Video Generation via In-Trajectory Resolution Scaling

Anonymous submission

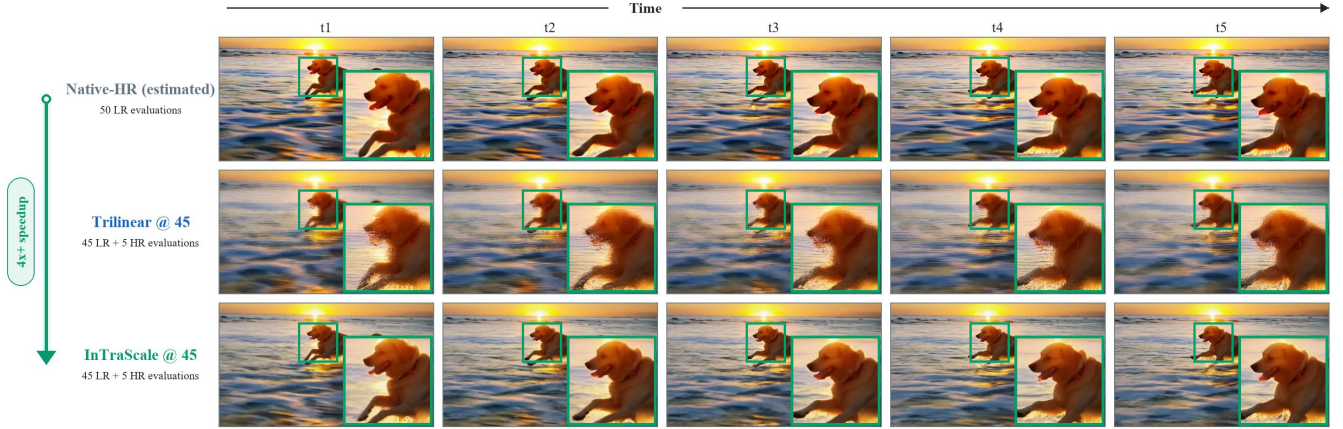


Figure 1: **Temporal comparison in decoded video space.** Five matched frames compare Native-HR (estimated), Trilinear@45, and InTraScale@45. The upsampled 368p Native-HR row is only a content-aligned reference, not the 720p quantitative baseline.

Abstract

High-resolution video diffusion remains expensive because every denoising step processes the full target-resolution spatiotemporal latent sequence. In-trajectory resolution scaling reduces this cost by executing an inexpensive low-resolution prefix and reserving target-resolution computation for the remaining steps, without launching a separate super-resolution diffusion process or increasing the total denoising-step budget. However, fixed trilinear interpolation cannot recover the VAE-induced cross-resolution latent correspondence, while a learned upsampler faces a training–inference mismatch between completed clean latents and predictions from an unfinished trajectory. We introduce InTraScale, a decoupled framework that addresses these challenges while keeping the pre-trained generator frozen. Its In-Trajectory Upsampler learns the cross-resolution correspondence from content-aligned clean latent pairs, whereas Trajectory-Tail Distillation distills the omitted low-resolution trajectory tail into a transition-local update, moving the transition clean prediction toward the completed low-resolution endpoint. A scheduler-consistent state reconstruction then returns the prediction to the remaining target-resolution trajectory. On a 50-step Wan2.1

sampler, InTraScale achieves $3.21\times$ and $4.44\times$ warm-model speedups while retaining 97.76% and 97.53% of the Native-HR VBench-5 score, respectively. On a distilled four-step model, it achieves a $2.13\times$ speedup over Native-HR4 while retaining 99.58% of the native distilled score. These results establish in-trajectory resolution scaling as an effective spatial acceleration strategy complementary to timestep distillation.

1 Introduction

Recent video diffusion models generate increasingly coherent motion and fine-grained visual content, but their computational cost remains a major obstacle to practical deployment (Wan et al. 2025). Each denoising step applies a large diffusion transformer to the complete spatiotemporal latent sequence, making target-resolution generation particularly expensive. Timestep distillation reduces the number of steps, but every retained step still processes the full target-resolution grid. Spatial acceleration is therefore complementary to step-count reduction: global layout, semantic structure, and motion can be established economically at low resolution, while expensive target-resolution computation is reserved for recovering local structure and high-frequency details.

Such low-to-high-resolution pipelines can follow two schedules, depending on whether resolution is increased af-

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ter low-resolution generation or before its trajectory is complete. Most existing pipelines follow the former, completing a low-resolution trajectory before applying an RGB-space or latent-space upsampling stage (Ho et al. 2022; Blattmann et al. 2023; Wang et al. 2023b). RGB-space cascades incur decode–encode and scheduler-state reconstruction overhead, whereas latent cascades directly upsample the completed representation (Xie et al. 2025; Zhao et al. 2026). Both, however, perform resolution enhancement only after low-resolution generation has completed. In-trajectory resolution scaling follows the latter: it transitions earlier and uses the remaining denoising steps as the target-resolution continuation (Jeong et al. 2025; ModelTC 2026), avoiding a separate super-resolution diffusion process or an increase in the original step budget.

Moving the resolution transition inside the denoising trajectory, however, creates two coupled representation challenges. Simple latent interpolation only resamples the spatial grid and cannot recover the cross-resolution correspondence induced by the pretrained VAE. The resulting structural distortion becomes particularly difficult to correct under a late transition, where only a short target-resolution continuation remains. A natural alternative is to replace interpolation with a learned latent upsampler. However, the upsampler is typically trained using completed, content-aligned clean latent pairs, whereas in-trajectory inference provides a clean prediction derived from an unfinished low-resolution trajectory. This prediction retains a timestep-dependent residual that the omitted low-resolution trajectory tail would otherwise reduce. Directly applying the learned upsampler therefore forces it to perform both cross-resolution reconstruction and unfinished trajectory correction, creating a training–inference mismatch. These two functions should consequently be modeled separately.

To address these problems, we introduce **InTraScale**, a decoupled framework for learned in-trajectory resolution scaling. The **In-Trajectory Upsampler (ITU)** learns the VAE-specific correspondence between completed clean low- and target-resolution latent pairs, replacing geometry-only interpolation with learned cross-resolution reconstruction. **Trajectory-Tail Distillation (TTD)** separately distills the omitted low-resolution trajectory tail into a transition-local update, moving the transition clean prediction toward the completed low-resolution endpoint. InTraScale then reconstructs the next target-resolution scheduler state and executes the remaining steps using the frozen flow predictor. On conventional 50-step Wan2.1 sampling, InTraScale achieves $3.21\times$ and $4.44\times$ warm-model speedups while retaining 97.76% and 97.53% of the Native-HR VBench-5 score, respectively. Applied to a distilled four-step model, it achieves a $2.13\times$ speedup over Native-HR4 while retaining 99.58% of the native distilled-model score.

Our contributions are summarized as follows:

- We introduce InTraScale, a learned in-trajectory resolution-scaling framework that distributes a fixed denoising-step budget between an inexpensive low-resolution prefix and a target-resolution continuation.
- We decouple resolution conversion from unfinished tra-

jectory correction: ITU learns the VAE-specific cross-resolution clean-latent correspondence, while TTD distills the omitted low-resolution trajectory tail into a transition-local update that calibrates the transition clean prediction toward the completed low-resolution endpoint.

- We evaluate InTraScale on conventional 50-step and distilled four-step Wan2.1 samplers, demonstrating substantial spatial acceleration both before and after timestep distillation.

2 Related Work

Post-Generation Resolution Enhancement. Post-generation methods enhance a completed low-resolution output in RGB space (Zhou et al. 2024; Xu et al. 2024; He et al. 2024) or latent space (Xie et al. 2025; Zhao et al. 2026). MrFlow combines pixel-space super-resolution with subsequent high-resolution refinement for image flow-matching models (Zheng et al. 2026b), while FlashVSR distills diffusion-based restoration into a one-step streaming model (Zhuang et al. 2026). Despite different representations and inference costs, these methods operate after the low-resolution trajectory has completed, whereas InTraScale reduces target-resolution computation within the original generative trajectory.

Multiscale and Resolution-Changing Generation. Matryoshka Diffusion and Pyramidal Flow Matching train generators around multiscale denoising processes (Gu et al. 2024; Jin et al. 2025). For pretrained models, RALU and LightX2V prescribe variable-resolution inference schedules (Jeong et al. 2025; ModelTC 2026), while Fresco and Spectral Progressive Diffusion progressively expand noisy states using spatial or spectral constructions (Zheng et al. 2026a; Xiao et al. 2026). These approaches either incorporate multiscale generation during training or define how noisy latent states are expanded. InTraScale instead keeps the generator frozen, learns the VAE-specific correspondence between clean low- and target-resolution latents, and separately corrects the unfinished low-resolution trajectory tail.

Orthogonal Video Diffusion Acceleration. Dynamic-computation methods vary model budgets or patch sizes across denoising steps and spatial regions (Anagnostidis et al. 2025; Kim, Ghadiyaram, and Gadde 2026; Li et al. 2026). Step-count reduction decreases the number of denoising steps through consistency or transition distillation (Wang et al. 2023a; Li et al. 2024; Ding et al. 2025; Nie et al. 2026), whereas caching methods reuse attention maps or intermediate features (Zhao et al. 2025; Liu et al. 2025). DisCa further learns caching under a restricted distilled trajectory (Zou et al. 2026). InTraScale instead reduces the spatial cost of each retained step and is therefore complementary to timestep distillation and feature reuse, as demonstrated on a distilled four-step generator.

3 Method

This section develops InTraScale by decomposing the transition into cross-resolution reconstruction and unfinished-trajectory correction. Accordingly, ITU learns the clean-

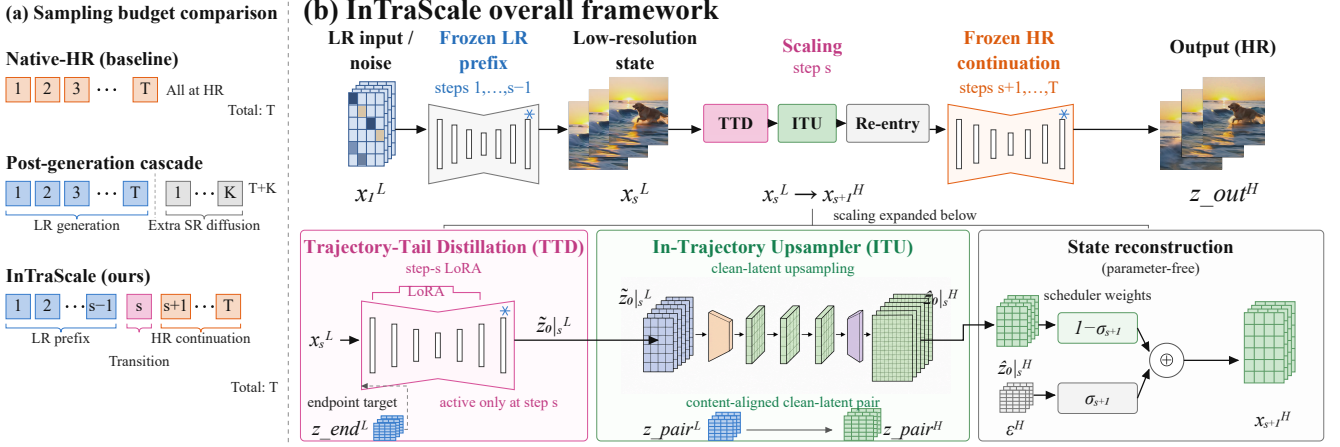


Figure 2: **Sampling budget and overall framework of InTraScale.** (a) Native-HR executes all T steps at target resolution, whereas a post-generation cascade appends K super-resolution steps after completing low-resolution generation. (b) At step s , TTD distills the omitted low-resolution trajectory tail into an endpoint-calibrated clean prediction. ITU, trained on content-aligned clean-latent pairs, lifts this prediction to the target-resolution latent space.

latent correspondence across resolutions, TTD calibrates the transition prediction toward the completed low-resolution endpoint, and parameter-free state reconstruction returns the lifted latent to the target-resolution trajectory.

3.1 Challenges of In-Trajectory Transition

Latent distortion from trilinear interpolation. A fixed trilinear interpolation operator only resamples the spatial grid; it does not recover the correspondence between the low- and target-resolution representations induced by the pre-trained VAE. Consequently, interpolated latents may exhibit distorted structures and inaccurate high-frequency statistics on the target-resolution grid. RALU reported that late latent upsampling introduces aliasing artifacts, particularly in high-frequency regions around object boundaries (Jeong et al. 2025). This exposes a fundamental quality–efficiency trade-off: a later resolution transition provides greater acceleration but leaves fewer target-resolution steps to correct the errors introduced during resolution conversion. With only a short target-resolution suffix, the remaining target-resolution steps may therefore be insufficient to repair structural distortions and synthesize details absent from the low-resolution representation.

Training–inference mismatch for learned upsampling.

Replacing interpolation with a learned latent upsampler introduces an input distribution shift. A latent upsampler is naturally trained using paired clean latents obtained from completed, content-aligned low- and target-resolution videos. Its low-resolution training input is thus a completed clean representation containing all spatial and temporal details supported by the low-resolution bandwidth. During in-trajectory inference, however, the upsampler does not receive such a converged latent. Let x_s^L denote the input to transition step s , the last step executed at low resolution; steps $s+1, \dots, T$ run at target resolution. Here, v_ϕ is the frozen flow predictor with parameters ϕ , c is the text condition, and σ_s is the

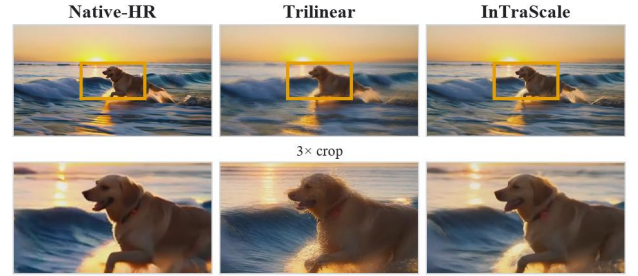


Figure 3: Trilinear interpolation under a late resolution transition. The columns show content-aligned Native-HR, trilinear interpolation, and final decoded InTraScale outputs obtained with a transition at step 45.

scheduler coefficient. We use the subscript $0|s$ for the clean latent predicted at the transition step. The upsampler therefore receives a transition clean prediction derived from an unfinished low-resolution trajectory:

$$\hat{z}_{0|s}^L = x_s^L - \sigma_s v_\phi(x_s^L, s, c).$$

It contains a timestep- and scheduler-dependent residual that would otherwise be reduced by the remaining low-resolution steps. Its distribution therefore differs from that of the clean low-resolution latents used to train the upsampler. Directly applying the upsampler to this prediction forces a single module to perform both cross-resolution reconstruction and unfinished trajectory correction, weakening its ability to preserve structure across scales.

These two challenges motivate the decoupled design of InTraScale. The In-Trajectory Upsampler learns the cross-resolution clean-latent correspondence, while Trajectory-Tail Distillation distills the omitted low-resolution trajectory tail into a transition-local update that moves the transition clean prediction toward the completed low-resolution endpoint.

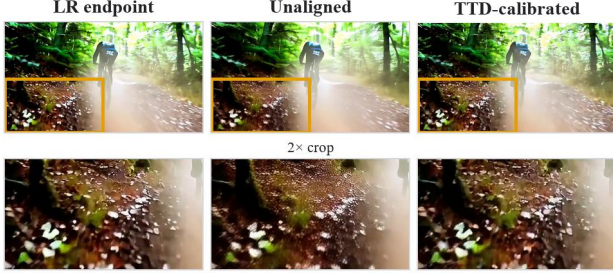


Figure 4: Unfinished-trajectory residual at transition step 45, the last step executed at low resolution.

3.2 Overall Framework

Figure 2 summarizes the sampling budget and the decoupled pipeline motivated by the two challenges above. We now formalize its components and trajectory states. Let $v_\phi(x_k^R, k, c)$ denote the frozen flow predictor, where $R \in \{L, H\}$ indicates the low- or target-resolution latent space, x_k^R is the latent state presented at denoising step k , and c is the text condition. We use “step” to denote one scheduler update and its corresponding model evaluation, σ_k for the scheduler coefficient, and T for the total number of denoising steps. We refer to the pretrained flow predictor and VAE collectively as the frozen base generator. Following the convention above, step s is the transition step and the last step executed on the low-resolution grid. Steps $1, \dots, s-1$ form the frozen low-resolution prefix, the TTD update is applied only at step s , and steps $s+1, \dots, T$ form the frozen target-resolution continuation.

At the transition step, the prediction $\hat{z}_{0|s}^L$ introduced in the preceding subsection is not directly passed to the upsampler. Trajectory-Tail Distillation (TTD) first produces an endpoint-calibrated clean prediction $\hat{z}_{0|s}^L$. The In-Trajectory Upsampler (ITU), parameterized by ψ , then maps this prediction to the target-resolution clean latent $\hat{z}_{0|s}^H = U_\psi(\hat{z}_{0|s}^L)$.

To continue the original denoising trajectory, InTraScale reconstructs the next target-resolution scheduler state as

$$x_{s+1}^H = (1 - \sigma_{s+1})\hat{z}_{0|s}^H + \sigma_{s+1}\epsilon^H(\xi), \quad \epsilon^H(\xi) \sim \mathcal{N}(0, I),$$

where $\epsilon^H(\xi)$ is target-grid Gaussian noise deterministically drawn from generation seed ξ . This parameter-free operation constructs a target-resolution state at scheduler level $s+1$, without auxiliary diffusion or a decode–encode round trip. We require $s < T$, so the frozen predictor performs at least one target-resolution step. The remaining target-resolution steps refine structural inconsistencies and synthesize details underdetermined by the low-resolution latent. The complete inference pipeline is

$$x_1^L \xrightarrow{\text{LR}} x_s^L \xrightarrow{\text{TTD}} \hat{z}_{0|s}^L \xrightarrow{\text{ITU}} \hat{z}_{0|s}^H \xrightarrow{\text{reconstruction}} x_{s+1}^H \xrightarrow{\text{HR}} z_{\text{out}}^H.$$

Here, z_{out}^H denotes the final target-resolution clean latent, which is decoded into the output video by the frozen VAE.

The ITU and TTD parameters are trained independently using supervision generated by the frozen base generator.

ITU learns the cross-resolution clean-latent correspondence, whereas the TTD update learns to reduce the unfinished low-resolution trajectory residual at the transition step.

3.3 In-Trajectory Upsampler

The In-Trajectory Upsampler (ITU) addresses the cross-resolution representation mismatch identified in Section 3.1. Rather than treating latent upsampling as spatial interpolation, ITU learns the correspondence between clean low- and target-resolution representations induced by the frozen base generator.

Model-endogenous supervision. We construct the training pairs using videos generated by the frozen base generator. Given a generated target-resolution video v^H , we first obtain its content-aligned low-resolution counterpart through RGB-space downsampling:

$$v^L = \mathcal{D}_{\text{rgb}}(v^H).$$

The two videos are then encoded independently by the same frozen VAE $\mathcal{E}$:

$$z_{\text{pair}}^L = \mathcal{E}(v^L), \quad z_{\text{pair}}^H = \mathcal{E}(v^H).$$

The resulting pair $(z_{\text{pair}}^L, z_{\text{pair}}^H)$ shares the same semantics, frame count, and motion trajectory. Performing the resolution change before VAE encoding allows ITU to learn the base generator’s actual cross-resolution latent correspondence instead of imitating a predefined interpolation operator in latent space.

Spatiotemporal latent upsampling. ITU maps a clean low-resolution latent to the target-resolution latent grid while preserving the temporal dimension:

$$U_\psi : \mathbb{R}^{B \times 16 \times F \times h_L \times w_L} \rightarrow \mathbb{R}^{B \times 16 \times F \times h_H \times w_H},$$

where B and F denote the batch size and number of latent frames, respectively. The network follows an encoder–resampler–decoder design. A 3D convolution first projects the input into a 256-channel hidden representation, followed by four spatiotemporal residual blocks. A learned spatial lifting operator then changes only the height and width. Four additional residual blocks and a final 3D convolution reconstruct the 16-channel target latent. The 3D blocks allow adjacent latent frames to contribute to the reconstruction without changing the temporal resolution.

We instantiate the spatial lifting operator according to the required grid ratio. For $480 \times 832 \rightarrow 720 \times 1248$, the latent grid changes from 60×104 to 90×156 . We realize this $3/2$ ratio using a $3 \times$ spatial PixelShuffle followed by anti-aliased stride-2 downsampling. For $368 \times 640 \rightarrow 720 \times 1248$, a $2 \times$ PixelShuffle produces a 92×160 grid, which is center-cropped to 90×156 . The resolution-specific lifting operators determine the target geometry, while the surrounding spatiotemporal network learns the latent correspondence.

Cross-resolution objective. For $\hat{z}_{\text{pair}}^H = U_\psi(\hat{z}_{\text{pair}}^L)$, the ITU objective is

$$\mathcal{L}_{\text{ITU}} = \mathcal{L}_{\text{rec}} + 0.2\mathcal{L}_{\text{lf}} + 0.1\mathcal{L}_{\text{temp}}.$$

Here, $\mathcal{L}_{\text{rec}}$ is the Charbonnier reconstruction loss, $\mathcal{L}_{\text{lf}}$ enforces consistency between the spatially downsampled prediction and the low-resolution input, and $\mathcal{L}_{\text{temp}}$ matches first-order temporal differences between adjacent latent frames. We use a Charbonnier constant of $\epsilon = 10^{-3}$. Here, $\mathcal{D}_{\text{lat}}$ maps the prediction to the low-resolution grid, and Δ_t denotes first-order temporal differences. The three terms constrain target-latent reconstruction, low-frequency content preservation, and temporal evolution, respectively.

A low-resolution latent does not uniquely determine every high-frequency detail in its target-resolution counterpart. ITU therefore focuses on transferring structure, motion, and latent statistics to the target grid. The remaining frozen target-resolution trajectory retains the generative responsibility for details that are underdetermined by the low-resolution input.

3.4 Trajectory-Tail Distillation

ITU is trained on completed clean-latent pairs, whereas its inference input is predicted from an unfinished low-resolution trajectory. TTD reduces this training–inference mismatch by distilling the omitted low-resolution trajectory tail into a transition-local update. The resulting endpoint-calibrated prediction is subsequently lifted by ITU and returned to the target-resolution continuation.

For endpoint supervision, we run the frozen base generator for each training sample using a fixed prompt, generation seed, initial noise, scheduler, and guidance configuration. We cache the state x_s^L immediately before the transition step and continue the same low-resolution trajectory to completion. We denote the resulting clean endpoint by z_{end}^L . The pair

$$(x_s^L, z_{\text{end}}^L)$$

therefore preserves the semantic and stochastic conditions at the transition step while providing direct supervision for the trajectory residual remaining at step s .

We implement TTD through a transition-specific low-rank update $\Delta\theta_s$ to the frozen flow predictor. For a linear layer W , the LoRA parameterization is

$$W_\lambda = W + \lambda \frac{\alpha}{r} BA,$$

where r is the LoRA rank, α is its training scale, and λ controls the TTD update strength during inference. We use rank $r = 32$ and inject the update into the q , k , v , and o attention projections and the two feed-forward projections. The base parameters ϕ remain frozen.

At step s , the TTD-augmented predictor produces

$$\tilde{z}_{0|s}^L = x_s^L - \sigma_s v_{\phi+\lambda\Delta\theta_s}(x_s^L, s, c).$$

We train the TTD parameters to regress the corresponding completed low-resolution endpoint using

$$\mathcal{L}_{\text{TTD}} = \mathcal{L}_1 + 0.1\mathcal{L}_{\text{MSE}} + \lambda_{\text{tt}}\mathcal{L}_{\text{temp}}.$$

Here, $\mathcal{L}_1$ and $\mathcal{L}_{\text{MSE}}$ compare $\tilde{z}_{0|s}^L$ with z_{end}^L , while $\mathcal{L}_{\text{temp}}$ matches their first-order temporal differences. We set $\lambda_{\text{tt}} = 0.05$ for the earlier transition at step $s = 40$ in the 50-step sampler and zero for the later and distilled settings. We select

the inference strength $\lambda = 0.75$ on held-out development prompts and keep it fixed for all reported samples.

The TTD update is applied only at step s . Its strength is zero for the preceding steps, so the frozen low-resolution prefix and the cached pre-transition state x_s^L remain unchanged under matched deterministic conditions. TTD is therefore a transition-localized distillation update rather than a trajectory-wide step-count reduction model.

4 Experiments

We evaluate whether InTraScale improves the quality–efficiency trade-off under conventional 50-step sampling, remains effective with a four-step timestep-distilled sampler, and addresses the transition failures identified in Section 3.1. We report system-level comparisons followed by component and design analyses.

4.1 Experimental Setup

We evaluate InTraScale on both the standard 50-step Wan2.1 T2V-1.3B sampler and a four-step Wan2.1 T2V-14B StepDistill-CfgDistill sampler. This section first examines the 50-step setting, where steps 40 and 45 serve as transition steps. We additionally evaluate trilinear interpolation with transition steps 40, 45, and 48 to characterize the effect of increasingly delayed transitions. All configurations generate 81-frame videos at 720×1248 resolution from matched text prompts, random seeds, initial noise, schedulers, and guidance settings.

We evaluate ten matched prompt–seed pairs using VBench. We report Subject Consistency, Background Consistency, Motion Smoothness, Aesthetic Quality, and Imaging Quality; Overall (VBench-5) is their unweighted mean, not an official VBench aggregate. Paired analyses use prompt–seed pairs as units.

We measure single-video warm-model end-to-end latency after one untimed warm-up, using five matched prompt–seed generations per resident process. CUDA-synchronized timing covers denoising, TTD, ITU, state reconstruction, VAE decoding, and output writing, but excludes runner construction and checkpoint loading.

4.2 Overall Quality–Efficiency Trade-off

We compare InTraScale against Native-HR and three accelerated alternatives. Native-HR executes all 50 steps at the target resolution. Trilinear directly resamples the latent grid at different transition steps. RALU-style combines trilinear transition with noise–timestep rescheduling at step 45. Finally, ITU-5HR (LUVE-style) completes all 50 steps at low resolution, applies ITU to the resulting endpoint, and performs five additional target-resolution refinement steps. Figure 5 visualizes the resulting operating points, while Table 1 reports their exact values.

InTraScale@40 reduces warm-model latency from 187.54 to 58.49 seconds, corresponding to a $3.21\times$ speedup, while retaining 97.76% of the Native-HR VBench-5 score. Delaying the transition to step 45 further reduces latency to 42.26 seconds, achieving a $4.44\times$ speedup and retaining 97.53% of the Native-HR score. Moving the transition from step 40 to

Method	LR/HR	Subj. Cons. $\uparrow$	Back. Cons. $\uparrow$	Motion $\uparrow$	Aesthetic $\uparrow$	Imaging $\uparrow$	Overall $\uparrow$	Latency (s) $\downarrow$	Speedup $\uparrow$
Native-HR	0/50	0.96140	0.96961	0.98477	0.59493	0.63112	0.82836	187.54 $\pm$ 0.26	1.00 $\times$
Trilinear@40	40/10	0.92179	0.95916	0.97542	0.55312	0.48110	0.77812	58.12 $\pm$ 0.20	3.23 $\times$
InTraScale@40	40/10	0.92713	0.95947	0.97975	0.59451	0.58829	0.80983	58.49 $\pm$ 0.16	3.21 $\times$
Trilinear@45	45/5	0.92263	0.95986	0.97532	0.52675	0.45425	0.76776	41.98$\pm$0.19	4.47$\times$
RALU-style@45	45/5	0.91940	0.95873	0.97222	0.56949	0.60216	0.80440	42.01 $\pm$ 0.19	4.46 $\times$
InTraScale@45	45/5	0.92359	0.95844	0.97957	0.58032	0.59768	0.80792	42.26 $\pm$ 0.20	4.44 $\times$
Trilinear@48	48/2	0.91995	0.96193	0.97552	0.49471	0.46833	0.76409	32.27$\pm$0.18	5.81$\times$
ITU-5HR (LUVE-style)	50/5	0.92543	0.95046	0.97542	0.57839	0.57395	0.80073	44.00 $\pm$ 0.27	4.26 $\times$

Table 1: Wan50 warm-model quality–efficiency. Bold marks the two best values in each metric; Overall denotes VBench-5.

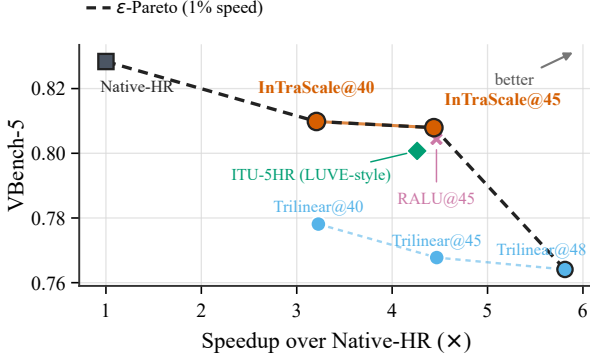


Figure 5: Wan50 quality–efficiency Pareto frontier.

45 reduces latency by an additional 27.74%, with a VBench-5 decrease of only 0.00191. The two operating points therefore provide a controllable quality–efficiency trade-off.

At matched transition positions, the learned transition introduces little additional latency over trilinear interpolation. InTraScale@40 and InTraScale@45 add only 0.63% and 0.67% latency, respectively, while improving VBench-5 by 0.03171 and 0.04016. The widening quality gap at the later transition supports the challenge identified in Section 3.1: as the target-resolution continuation becomes shorter, trilinear interpolation leaves insufficient steps to repair the induced latent distortion.

At step 45, InTraScale also improves VBench-5 over RALU-style rescheduling by 0.00352 with a 0.61% latency increase. This comparison suggests that RALU-style rescheduling combined with trilinear transition does not fully address the VAE-induced cross-resolution correspondence modeled by ITU.

The ITU-5HR (LUVE-style) baseline completes the entire low-resolution trajectory before resolution conversion and adds five target-resolution refinements. Despite executing these additional steps, it reaches only 0.80073 VBench-5 at 44.00 seconds. InTraScale@45 is 3.96% faster and improves VBench-5 by 0.00719. Transitioning before the low-resolution endpoint therefore provides a more favorable operating point than completing low-resolution generation and subsequently launching a separate target-resolution refinement suffix.

Overall, both InTraScale settings are favorable non-

dominated operating points on the evaluated tolerance-aware frontier. Their paired confidence intervals against Native-HR overlap zero, but this is not an equivalence test; we therefore do not claim lossless acceleration.

4.3 Further Accelerating Distilled Video Generation

Although Section 4.2 demonstrates substantial acceleration for conventional 50-step sampling, deployment-oriented video generators increasingly rely on timestep-distilled models to reduce the trajectory length. We therefore ask whether in-trajectory resolution scaling can provide additional gains after the step count has already been compressed. To this end, we apply InTraScale to a four-step Wan2.1 T2V-14B StepDistill-CfgDistill model. Step 3 is the transition step and the last step executed at low resolution, leaving one target-resolution step for generative refinement.

We compare against Native-HR4, which executes all four steps at the target resolution, and Trilinear@3, which uses the same 3-LR/1-HR schedule as InTraScale but replaces ITU with fixed trilinear interpolation and disables TTD. We further construct two controlled post-generation baselines on the same distilled backbone. Both complete all four low-resolution steps before increasing the spatial resolution. LUVE-style then applies learned latent upsampling followed by two target-resolution refinements, whereas MrFlow-style uses a decode–RGB-upsample–re-encode path followed by one target-resolution correction. These configurations reproduce the corresponding computational patterns rather than the complete original systems. Figure 6 visualizes these operating points, while Table 2 reports their exact values.

InTraScale-D4@3 reduces warm-model latency from 42.49 to 19.92 seconds, achieving a 2.13 $\times$ speedup while retaining 99.58% of Native-HR4 VBench-5. Its paired confidence interval overlaps zero, we therefore describe this difference as marginal rather than claiming quality equivalence.

Trilinear@3 reaches a slightly higher 2.32 $\times$ speedup, but its VBench-5 drops by 0.04244 relative to Native-HR4. At the same 3-LR/1-HR schedule, InTraScale adds only 1.59 seconds over Trilinear@3 while improving every component and Overall by 0.03884, driven by imaging quality (+0.12635) and aesthetic quality (+0.05960). This result strengthens the observation from the 50-step experiments: when the target-resolution continuation is reduced to a single step, trilinear interpolation leaves insufficient correction capacity, whereas the learned transition preserves most of

Method	LR/HR	Subj. Cons. $\uparrow$	Back. Cons. $\uparrow$	Motion $\uparrow$	Aesthetic $\uparrow$	Imaging $\uparrow$	Overall $\uparrow$	Latency (s) $\downarrow$	Speedup $\uparrow$
Native-HR4	0/4	0.97186	0.97557	0.99123	0.65203	0.71135	0.86041	42.49	1.00 $\times$
Trilinear@3	3/1	0.95932	0.96937	0.98376	0.58954	0.58783	0.81796	18.33	2.32$\times$
InTraScale-D4@3	3/1	0.96226	0.97026	0.98815	0.64914	0.71418	0.85680	19.92	2.13$\times$
LUVe-style	4/2	0.96238	0.97341	0.98683	0.65191	0.72450	0.85981	28.76	1.48 $\times$
MrFlow-style	4/1	0.96379	0.97259	0.98796	0.65491	0.71591	0.85903	27.12	1.57 $\times$

Table 2: Distill4 warm-model quality–efficiency. Bold marks the two best values in each metric; Overall denotes VBench-5.

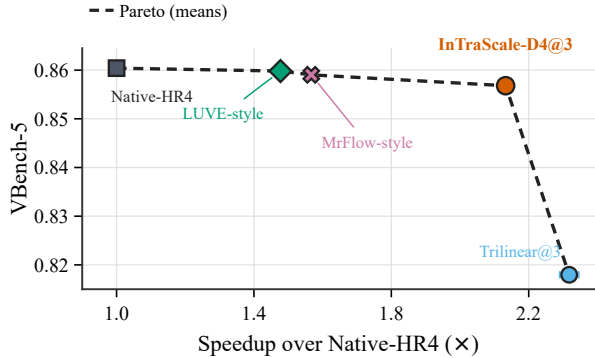


Figure 6: Distill4 quality–efficiency Pareto frontier.

the native quality.

InTraScale also provides a more favorable operating point than the post-generation alternatives. It reduces latency by 30.73% relative to LUVe-style and by 26.53% relative to MrFlow-style, while its VBench-5 is lower by only 0.00301 and 0.00223, respectively. The quality difference from MrFlow-style has a prompt-paired confidence interval that overlaps zero; the LUVe-style advantage is small but detectable. InTraScale additionally obtains higher motion smoothness than both post-generation baselines shown in Table 2.

These results demonstrate that in-trajectory resolution scaling is complementary to timestep distillation. Even after the denoising trajectory has been compressed to four steps, InTraScale reduces warm-model end-to-end latency by 53.1% while retaining 99.58% of the Native-HR4 VBench-5 score.

4.4 Component and Design Analysis

We next examine whether ITU and TTD address the two transition-specific challenges identified in Section 3.1. On 50 completed clean-latent pairs per scale, ITU consistently improves cross-resolution reconstruction over trilinear interpolation. We compute latent L_1 in latent space and LPIPS/temporal L_1 on decoded RGB frames, with the latter comparing first-order frame differences. For 480p $\rightarrow$ 720p, ITU reduces these three metrics by 48.6%, 73.3%, and 30.8%, respectively. Under the more aggressive 368p $\rightarrow$ 720p conversion, the corresponding reductions are 33.4%, 51.1%, and 15.5%. These results confirm that ITU learns the VAE-specific cross-resolution correspondence rather than

merely approximating geometric interpolation.

TTD consistently brings the transition clean prediction closer to the endpoint of the complete low-resolution trajectory. It reduces endpoint latent L_1 by 25.82%, 21.03%, and 5.03% at the 50-step transitions $s = 40$ and $s = 45$, and the D4 transition $s = 3$, respectively, with improvements on all ten samples in each setting. The factorial evaluation further separates the roles of the two modules: replacing trilinear interpolation with ITU improves VBench-5 by 0.03085–0.04066, whereas enabling TTD changes the aggregate score by only -0.00050 to $+0.00087$. ITU therefore provides the dominant perceptual gain, while TTD primarily reduces the transition-specific endpoint mismatch.

Finally, we validate the target-resolution state reconstruction on an independent eight-prompt set. Using seed-derived target-resolution noise instead of resized low-resolution flow improves VBench-5 by 0.02483 and temporal consistency by 0.01605, showing the importance of reconstructing a scheduler-consistent high-resolution state. Performance remains stable when the TTD update strength varies from 0.25 to 1.0; we select $\lambda = 0.75$ on the validation set and keep it fixed for all test evaluations. Full metric breakdowns and strength sweeps are provided in the supplementary material.

5 Conclusion

We presented InTraScale, which accelerates high-resolution video generation by scaling resolution within a fixed denoising budget, without a separate super-resolution diffusion process. TTD adapts the transition state, while ITU performs VAE-specific cross-resolution latent lifting, allowing the frozen generator to continue at the target resolution. On 50-step Wan2.1 sampling, InTraScale achieves 3.21 $\times$ and 4.44 $\times$ warm-model speedups while retaining 97.76% and 97.53% of the Native-HR VBench-5 score. On a four-step distilled model, it achieves 2.13 $\times$ acceleration with 99.58% score retention. Ablations confirm the complementary roles of TTD and ITU, establishing in-trajectory resolution scaling as a spatial acceleration strategy complementary to timestep distillation.

Limitations. Model-endogenous self-distillation specializes ITU to the VAE and resolution pair and TTD to the transition step. This enables close adaptation without external paired data, but new configurations may require renewed supervision and module adaptation. Future work will explore cross-configuration transfer and content-aware transition policies.

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